

PQC H-365

High-Performance Quantum-Resistant Hybrid VPN & Multi-Message Secure Channel

NIST FIPS 203 (ML-KEM-768) + NIST FIPS 204 (ML-DSA-65) with Asynchronous Parallel Handshake Acceleration and 64-bit Monotonic Replay-Protected Multi-Message Transport.

TEAM NAME: PQC H-365

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Track: Quantum Security & Cryptography

Problem Statement title: PQC H-365 Hybrid Quantum VPN

Repository: <https://github.com/chipashacosmas/PQC-H365>

Problem: What Problem Are You Solving?

The Quantum Break

- Shor's Algorithm executed on a Cryptanalytically Relevant Quantum Computer (CRQC) breaks all classical public-key cryptography.
- RSA-2048, Diffie-Hellman, and ECDH (P-256 / X25519) will be solved in polynomial time.
- Modern enterprise and defense VPN tunnels will provide zero forward secrecy.

Harvest Now, Decrypt Later

- Hostile nation-states and adversaries are actively recording encrypted network traffic today.
- Encrypted enterprise secrets, intellectual property, and government records are retained in bulk.
- Once quantum hardware arrives, stored archives will be retroactively decrypted.
- Migration must happen now — before quantum computers mature.

The Engineering Bottleneck

- Post-quantum algorithms produce kilobyte-scale keys and signatures (1KB to 3KB vs 32B).
- Sequential hybrid handshakes introduce high CPU and network latency (50+ ms stalls).
- Packets exceed standard network MTU (1500B), causing fragmentation and packet loss.
- Industry needs a high-speed, line-rate PQC architecture.

Why It Matters: Who Faces This Problem & Relevance

Critical Sectors at Immediate Risk

- **Defense & Intelligence:** Government networks transmitting classified intelligence subject to foreign intercept programs.
- **Banking & Financial Infrastructure:** SWIFT inter-bank financial communications and long-term financial agreements.
- **Critical Infrastructure & Cloud VPNs:** Enterprise remote work gateways and cloud interconnects defending customer data.

Significance of Immediate Action

- **Permanent Exposure Window:** High-value intelligence requires 20-30 years of confidentiality; waiting for quantum supremacy guarantees data breach.
- **Gateway Scalability:** Traditional serial handshakes (>110 ms round-trips) drop sessions and crash high-concurrency VPN concentrators.
- **Compliance Deadline:** Mandated by NSA Commercial National Security Algorithm Suite 2.0 (CNSA 2.0) timelines.

Proposed Solution: What Are You Building?

Dual-Security Hybrid Key Exchange

- Combines classical X25519 (ECDH) with NIST FIPS 203 (ML-KEM-768).
- Shared secrets are fused via RFC 5869 HKDF-SHA256 into a 32-byte session key.
- Dual-Security: Adversaries must break BOTH elliptic curves and lattice mathematics simultaneously.

Mutual Post-Quantum Auth (mPQC)

- Implements NIST FIPS 204 (ML-DSA-65) post-quantum digital signatures.
- Server and client sign the entire ephemeral handshake transcript.
- Eliminates active quantum Man-in-the-Middle (MitM) and quantum downgrade attacks.

Parallel Handshake Scheduling

- POSIX multithreaded asynchronous key exchange pipeline in native C.
- Computes classical X25519 and ML-KEM-768 concurrently.
- Reduces handshake latency by 81.5% on client and 79.1% on server (down to ~11ms).

Replay-Protected Secure Channel

- Persistent multi-message AES-256-GCM authenticated transport.
- 64-bit monotonic sequence numbers bound directly to AES-GCM nonces.
- Hardware wire-speed encryption with authenticated clean session teardown.

Why Quantum? Technology Rationale & Enabler



Mathematical Necessity

- The quantum threat is algorithmic: Shor's algorithm solves hidden subgroup problems on abelian groups in polynomial time.
- Classical mitigations (e.g., doubling RSA/ECC key size) offer zero protection against quantum speedup.
- Lattice-based cryptography (Module-LWE and Module-SIS) relies on worst-case shortest vector problems, currently immune to quantum polynomial solvers.



Enabling the Quantum-Safe Future

- **NIST Standardization:** Aligns with freshly finalized standards: NIST FIPS 203 (ML-KEM) and FIPS 204 (ML-DSA).
- **Real-World Enabler:** Connects theoretical quantum cryptography to real-world Linux network sockets with line-rate performance.
- **Zero Compromise:** Enables organizations to adopt quantum-safe communications without suffering classical latency regressions.

Implementation and Architecture: Technology Stack

C Cryptographic Engine

- **Native C Implementation:** High-performance, memory-safe C built with Open Quantum Safe (liboqs 0.10.0) and OpenSSL.
- **POSIX Worker Pool:** Thread pool concurrently runs X25519 key derivation and ML-KEM-768 encapsulation.
- **CMake & Ninja Build Pipeline:** Multi-target compilation for standalone KAT tests, benchmarking suite, and network daemons.

Framing & Network Architecture

- **Zero-Copy Adaptive Framing:** 4-byte network length prefix accommodates 1,184B KEM keys and 3,309B DSA signatures up to 4,096B with strict bounds checking.
- **Replay Defense:** 64-bit monotonic sequence counter bound to AES-256-GCM nonce and AAD ("PQC_H365_CHANNEL_V1").
- **Linux Network Integration:** Tested and verified on Ubuntu Server across isolated virtual network interfaces.

Minimal Prototype: What Have You Actually Built?

Complete Open Source Repository

- Public GitHub repository: <https://github.com/chipashacosmas/PQC-H365>
- **Core Library:** `libpqc_common.a` providing AEAD, framing, HKDF, timing, and state machine routines.
- **Daemon Binaries:** `pqc_channel_server` and `pqc_channel_client` executing live post-quantum tunnels.
- **Automated Test Suite:** `./scripts/run_all_tests.sh` with 100% passing tests.

Live Channel Execution Verification

- **Handshake:** Server ML-DSA-65 signature verified: `[CLIENT] server sig: PASS`.
- **Session Key:** Both derive identical SHA-256 session key prefix: `59b7b6397a0ea3bd`.
- **Message 1 (ctr=1):** Encrypted with AES-GCM; server echoes `ECH0: hello PQC SECURE CHANNEL`.
- **Message 2 (ctr=2):** Monotonic counter increment verified; message decrypted and echoed.
- **Teardown (ctr=3):** Clean shutdown triggered via `QUIT`; session closed gracefully.

Results / Validation: Findings & Benchmark Evidence

Handshake Mode	Client Avg (ms)	Server Avg (ms)	Total Latency (ms)	Quantitative Impact
Classical X25519 Baseline	1.558 ms	0.547 ms	2.105 ms	Baseline (Insecure vs Quantum)
ML-KEM-768 Standalone	8.432 ms	7.225 ms	15.657 ms	PQC Only (No Classical Fallback)
Sequential Hybrid	55.559 ms	56.314 ms	111.873 ms	Serial Overhead (Slow)
Parallel Hybrid (PQC H-365)	10.268 ms	11.789 ms	22.057 ms	81.5% Client / 79.1% Server Reduction

VALIDATED BREAKTHROUGH

81.5% Latency Cut

Proves post-quantum cryptographic security is achievable with ~11 ms latency over real TCP network sockets.

✓ 100% Verification Suite Pass

- Standalone KATs, 4096B framing boundaries, and replay defense fully verified with empirical datasets published to GitHub.

Roadmap: Round 2 Sprint & Grand Finale

Round 2 Enhancement Sprint (Oct 10 – 29)

- **Linux TUN/TAP Driver (Step 29):** Binding `tun0` virtual interface to route real OS-level IPv4/IPv6 packet flows.
- **eBPF / XDP In-Kernel Acceleration:** Implementing bare-metal packet filtering and line-rate decryption hooks.
- **Multi-Path Interface Bonding:** AEAD frame multiplexing across dual network physical interfaces for link failover.
- **Adversarial Fuzzing:** Scapy/Kali fuzzing suite validating resilience against malformed quantum frames.

Grand Finale (Bengaluru: Oct 30 – 31)

- **Live Hardware Demonstration:** Operating PQC H-365 tunnel on physical machines with Wireshark packet analysis.
- **Live Quantum MitM Attack Simulation:** Demonstrating active quantum stripping rejection and signature failure.
- **Telemetry Control Plane:** Demonstrating live latency charts and active client tunnel telemetry.
- **Open Source Release:** Packaged deployment binaries for Ubuntu / Debian and Docker containers.

References and Sources

- **NIST FIPS 203:** Module-Lattice-Based Key-Encapsulation Mechanism Standard (ML-KEM), National Institute of Standards and Technology, 2024.
- **NIST FIPS 204:** Module-Lattice-Based Digital Signature Standard (ML-DSA), National Institute of Standards and Technology, 2024.
- **RFC 5869:** HMAC-based Extract-and-Expand Key Derivation Function (HKDF), Internet Engineering Task Force.
- **RFC 7748:** Elliptic Curves for Security (X25519 & X448), Internet Engineering Task Force.
- **NIST SP 800-38D:** Recommendation for Block Cipher Modes of Operation: Galois/Counter Mode (GCM).
- **Open Quantum Safe (OQS):** liboqs C library for quantum-resistant cryptographic algorithms (<https://github.com/open-quantum-safe/liboqs>).
- **Original Team Repository:** PQC H-365 implementation, benchmarks, and documentation (<https://github.com/chipashacosmas/PQC-H365>).